

A schema for the LULC model & dataset zoo

Model cards, dataset cards, and the catalogue that ties them together

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Where we are, and the direction

Done so far

- A 4-class base map (greenery / water / built-up / barren) from Alpha Earth, at 10 m.
- A **living hierarchy**: split / add a class at any level, retrain on the fly, and the map updates, with metrics.

The direction (what was advised)

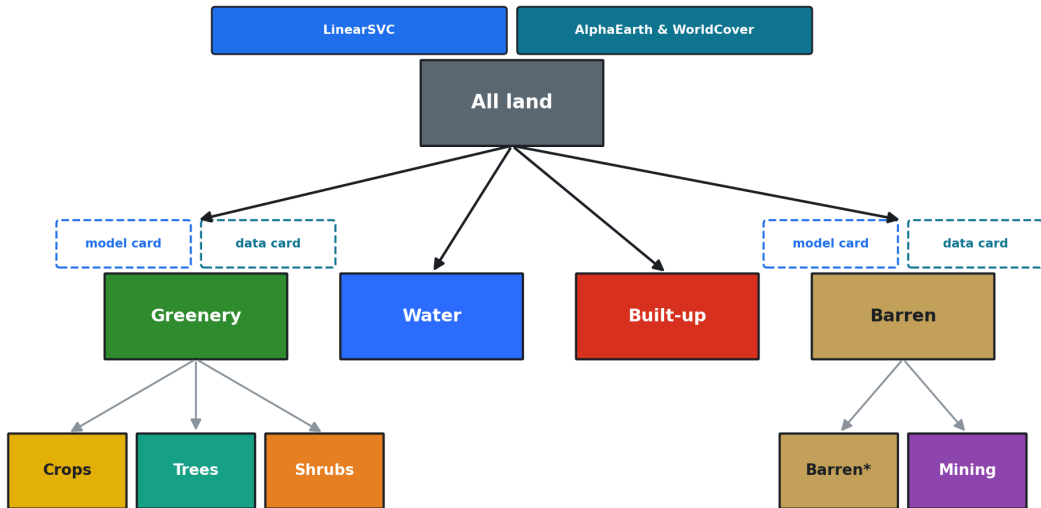
- build a **model zoo + dataset catalogue** with design.
- Each model tagged with *where it's valid*, what it emits, what it trained on, and how classes were annotated. We call these “model cards”.
- Let users browse, pick one for their area, and keep refining.

The structure: two record types (+ a spine)

What the schema defines is the *shape* of each record, not any one model:

- **Dataset Card:** a labeled *source* (polygons, an EE asset, or a pixel table) with a description, spatial/temporal extent, provenance/evidence, and quality.
- **Model Card:** a *classifier* at one node, holding the classes it emits, the datasets it trained on, where it's valid, its metrics, and how it deploys.
- **Canonical taxonomy:** the 4 base classes the cards hang off.
- **Catalogue:** a flat JSON registry of cards plus an index, queried to pick a model for an area.

The class tree the cards hang off



Our own taxonomy, growable at any level

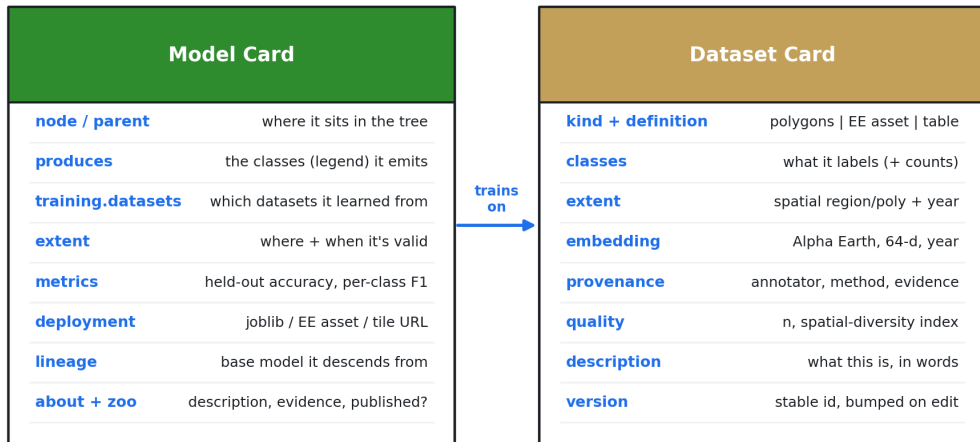
- Start from 4 base classes, but the tree grows at **any** level: split a class, add a class, or even **add a new base class** (which retrains the base map).
- New classes usually map **into** our taxonomy where they fit (a seasonal / perennial water-body under water), so models stay comparable; if something genuinely doesn't fit, it can become a new base class.
- Standard taxonomies (WorldCover / USDA / IUCN) aren't the backbone. They show up *only at retraining time*, as an optional cross-reference a user can attach.

Optional cross-reference (attached at retrain time):

our class	under	WorldCover	USDA
crops	greenery	Cropland (40)	Cropland
trees	greenery	Tree cover (10)	Forest land
mining	barren	Bare/sparse (60) *	Extractive

* WorldCover has no "mining", which is *why* you add it locally and roll it up to barren.

Anatomy of the two cards



Dataset Card: two ways to define a dataset

```
{ "id": "ds_farmforest_crops_v1", "kind": "polygons",
  "description": "expert-delineated cropland parcels (crop positives)", // describe it
  "classes": [ {"class":"crops","name":"Crops","count":110} ],
  "definition": { // (1) polygons OR (2) an EE asset
    "type": "polygons", "path": "data/examples/crops.geojson"
    // "type":"ee_asset", "asset":"ESA/WorldCover/v200", "band":"Map", "code":20
  },
  "extent": { "spatial": {"type":"region","value":"India"}, "temporal": {"year":2024} },
  "provenance": { "annotator":"FarmForest expert", "method":"expert delineation",
    "evidence":["drone imagery over IIT plots"] }, // how annotated + evidence
  "quality": { "n_polygons":110, "spatial_diversity":0.889 } } // diversity / counts
```

Model Card: a classifier at one node

```
{ "id": "mc_greenery_split_v1", "node": "greenery", "parent_class": "greenery",
  "topology": "per_node_split",
  "produces": [ {"class":"crops","std_mapping":{"worldcover":40}},
                 {"class":"trees"}, {"class":"shrubs"} ], // the legend it emits + map-out
  "training": { "datasets":["ds_farmforest_crops_v1","ds_farmforest_trees_v1",
                          "ds_worldcover_shrubs_v1"],
                "algo":"StandardScaler->LinearSVC",
                "balancing":{"method":"class_weight_balanced","residual_cap":8000} },
  "extent": { "spatial":{"type":"region","value":"India"}, "temporal":{"year":2024} },
  "metrics": { "accuracy":0.963, "macro_f1":0.884, "eval":"polygon-holdout" },
  "deployment": { "ee_asset":null, "tile_url":null, "expressible_as_bandmath":true },
  "lineage": { "base_model":"mc_base_pooled_v1", "derived_from":null },
  "about": { "description":"...", "intended_use":"...", "limitations":"shrubs weak" } }
```


The typed extent object (validity)

One shape, three spatial forms plus a temporal field, so human AEZ labels and precise geometries coexist. Used on *both* cards.

```
"spatial": {"type":"region","value":"India"} // world | AEZ-13 | district:...  
// {"type":"polygon","geojson":{...}} // a drawn / uploaded boundary  
// {"type":"ee_asset","asset":"users/.../aez_13"}  
"temporal": {"year":2024} // OR {"start":"2020","end":"2024"}
```

- “Is this area inside the model’s extent?” For a region, look up its geometry and test; for a polygon or EE asset, a direct geometry test.
- A dataset is valid *somewhere*; a model is valid *somewhere*; it’s the same object.
- Space and time aren’t the only axes: a model is also bound to the **embedding it was trained on** (Alpha Earth, with its version, year, and coverage). The object stays *open* to such axes; we start with the ones we can actually check.

Choosing the training data: the dataset panel

What the user picks from

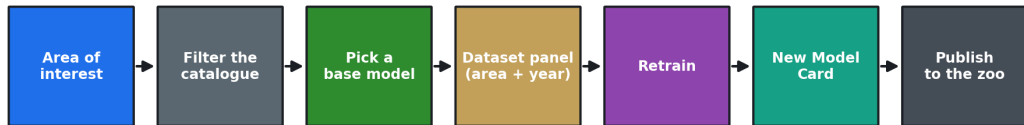
- **Their own** polygons (drawn / uploaded).
- A **curated standard library** of pre-written cards for ESA WorldCover slices and USDA / IUCN refs. *Offered, not web-searched* (provenance + reproducible).
- Tag each **positive** or **negative**; select / unselect freely.

Preferences applied to sampling

- **Spatial**: “only my area of interest” → clip to the AOI.
- **Temporal**: “only year X” → sets the sampling & embedding year.

Recorded on the model as `training.datasets` + a selection block (AOI + year) so a retrain is reproducible.

Pick, refine, publish



Quality & imbalance: guidelines baked in

- **Spatial diversity index** (dataset quality): Shannon entropy of sample locations over a coarse grid, normalized to $[0, 1]$. ≈ 1 = well spread; ≈ 0 = all from one spot. Flags “100 polygons, all one district”. Measured on our demo data: crops **0.889**, mining **0.867**, trees **0.840** (healthily spread).
- **Imbalance guidelines** (before a split / add trains):
 - compute the resulting class ratios; if skew passes a threshold (e.g. 5:1), warn;
 - offer a fix: *undersample* majority / *oversample* minority / cap the residual (already capped at 8000) / re-weight;
 - record the choice in `training.balancing`, and keep showing per-class metrics (a class can be present yet unlearnable, e.g. shrubs F1 0.695).

Ties to our finding: *diversity*, not raw volume, moves accuracy on unseen regions.

Worked examples (real numbers, today's models)

Model Card	node	topology	held-out acc.
mc_base_pooled_v1	root	base_pooled	0.80 / 0.89 [†]
mc_greenery_split_v1	greenery	per_node_split	0.963
mc_barren_mining_v1	barren	per_node_split	0.867

[†] random India / balanced expert hold-out.

- greenery: crops F1 0.965, trees 0.991, shrubs 0.695 (WorldCover-labelled, weak).
- mining (ADD = split + residual): mining F1 0.810 on 100 real polygons.
- Every field is filled from something we **already have**; for now the schema is **validated locally** against our existing models and datasets.

Thank you